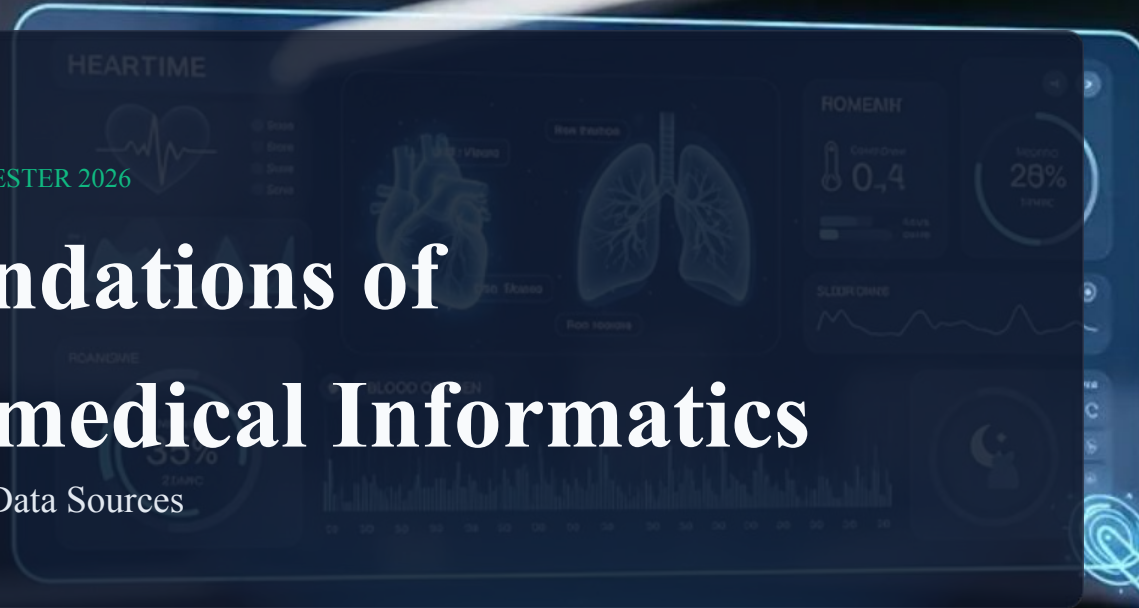


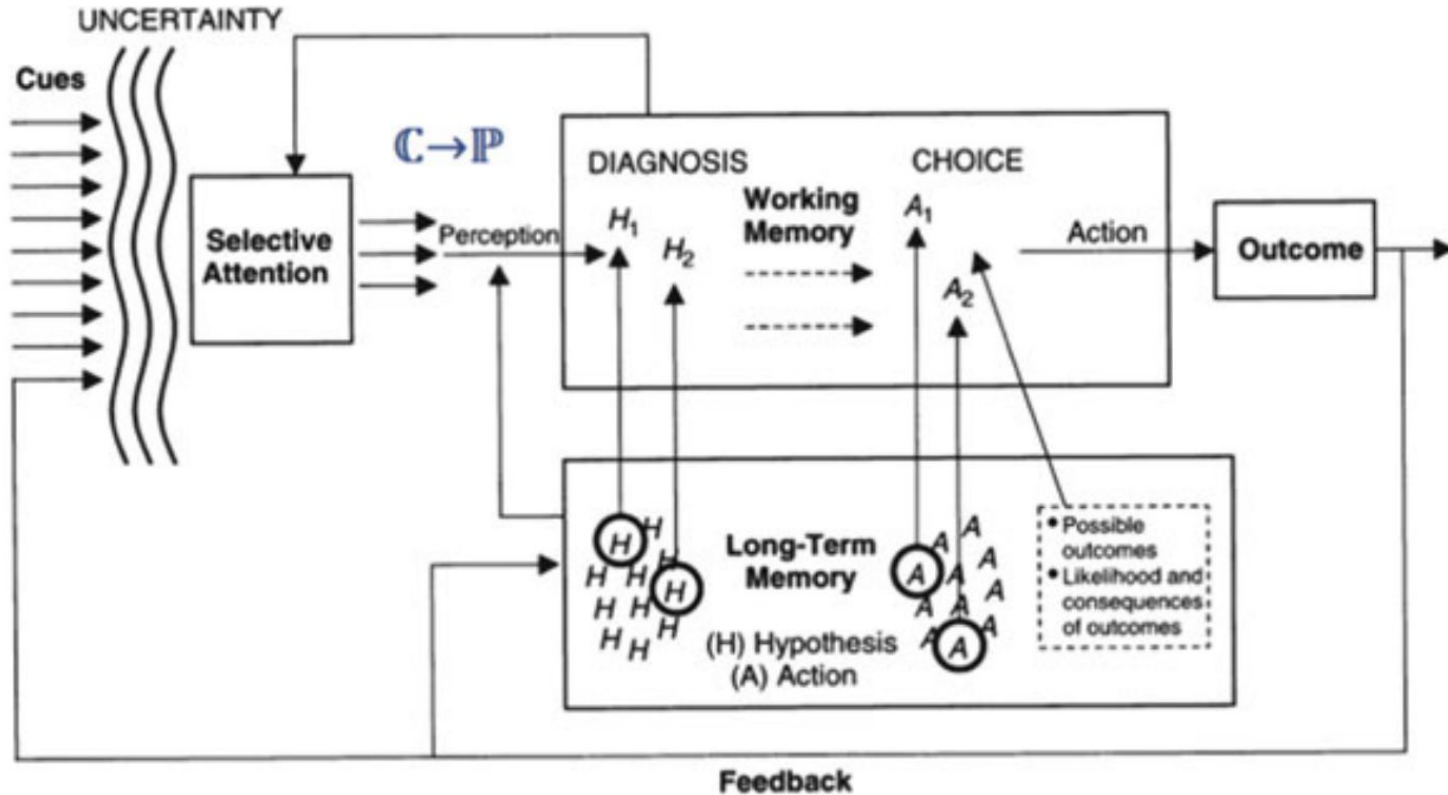
WINTER SEMESTER 2026

Foundations of Biomedical Informatics

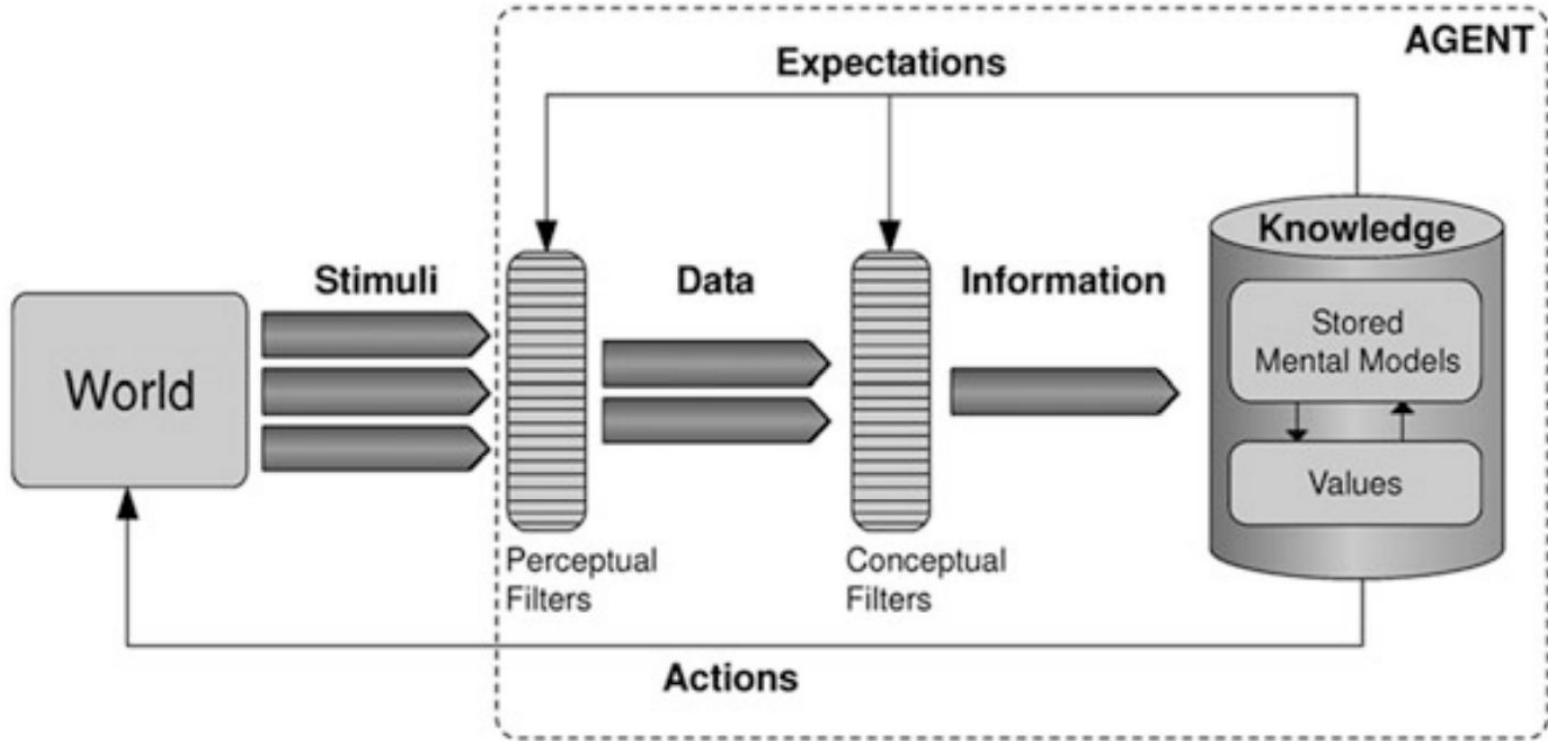
Lecture 4: Data Sources



A Cognitive Model of Decision Making in Clinical Scenarios



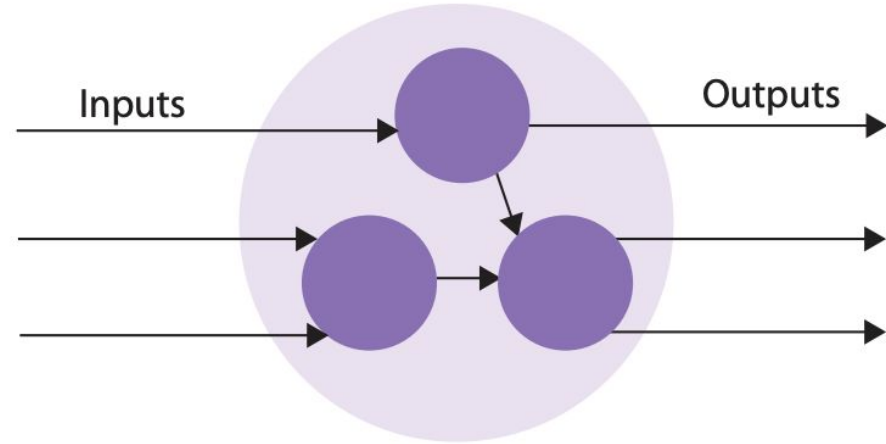
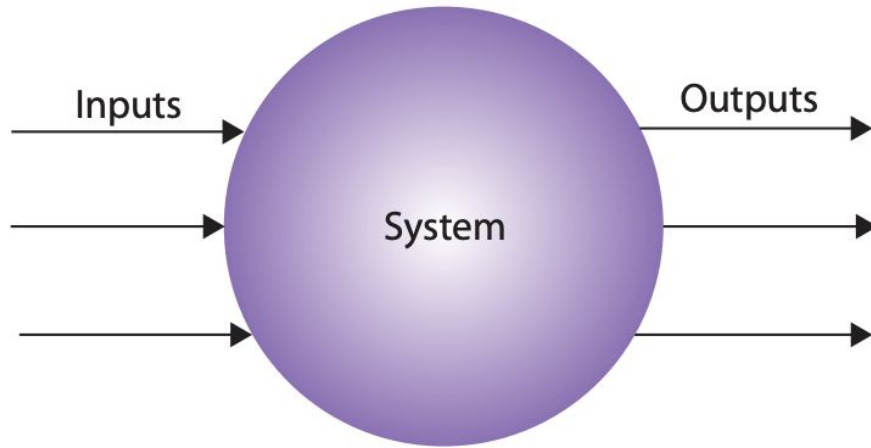
Human Information Processing Model



What is a Learning Health System Model?

- Routine care generates data → evidence → improved care (continuous feedback loop)
- Alignment needed
- science + informatics
- incentives
- culture
- governance/ethics
- Goal: best care at lower cost, continuously improving

Learning Health System



Learnings

- Expect **adaptation + feedback loops**
- Monitor early and continuously (surveillance + rapid response)
- Use multi-pronged strategies (combination therapy, stewardship, infection control)
- Update models with real-world data (continuous learning)
- Build policies assuming **unknown unknowns** (resilience, not certainty)

Bottom line: *Act, but design interventions as experiments in a complex system.*

How Systems Work: Example from Antibiotic Resistance

What happened

- Antibiotics created huge optimism: infectious disease looked “solvable.” Over time, bacteria evolved resistance (e.g., penicillinase producers).
- Antibiotics **selected for resistant survivors**, who then dominated populations.
- Some resistant infections now disrupt entire hospitals.

Why it happens (systems lens)

- Every intervention is based on a **simplified model** of reality. We act assuming “everything else is equal” → but systems never behave that cleanly. We rely on the **closed-world assumption**: “we know enough to act.”
- Resistance is not just a biology problem it’s a **predictable systems outcome** of intervention under **incomplete knowledge**.

Sources of Health Data

JOHNS HOPKINS
UNIVERSITY & MEDICINE

Welch Medical Library

[Welch Medical Library Home](#) / [Welch Medical Library Guides](#) / [Finding Datasets for Secondary Analysis](#) / [Research Data Repositories & Databases](#)

Finding Datasets for Secondary Analysis

About This Guide

New to Hopkins?

Research Data Repositories &
Databases

NIH Data Repositories

Examples of NIH Data
Repositories

Other Data
Repositories/Consortium

NIH Data Repositories

In general, NIH does not endorse or require sharing in any specific repository and encourages researchers to select the repository that is most appropriate for their data type and discipline (though such specification does exist for particular initiatives). To help researchers locate an appropriate resource for sharing their data, as well as to promote awareness of resources where datasets can be located for reuse, Trans_NIH BioMedical Informatics Coordinating Committee (BMIC) maintains lists of several types of data sharing resources:

- [Open NIH-supported domain-specific repositories](#) that house data of a specific type or related to a specific discipline;
- [Other NIH-supported domain-specific resources](#), including repositories and knowledgebases, that have limitations on submitting and/or accessing data; and
- [Generalist repositories](#) that house data regardless of type, format, content, or subject matter.

<https://browse.welch.jhmi.edu/datasets/nih-data-repositories>

Sources of Health Data

 **HARVARD**
Dataverse

Add Data ▾ Search ▾ About User Guide Support Sign Up Log In

The Repository is utilizing a "Submit for Review" workflow (please see [support page](#) for more information). Deposits will be reviewed within 24 hours, Monday - Friday from 9am to 5pm. All deposits after hours and on weekends will be reviewed the next business day. ✕

 Metrics

127,069,400 Downloads

 Contact  Share

Antibiotic Resistance  Advanced Search

+ Add Data

  **Dataverses (0)**

  **Datasets (5)**

  **Files (1)**

Publication Year
2026 (6) ✕

Subject
[Social Sciences \(3\)](#)
[Medicine, Health and Life Sciences \(1\)](#)
[Physics \(1\)](#)

Author Name
[Amy Watterson, Amanda Hubbard, Rui Diaz-Pacheco, Ashleigh Francis, Robert Granetz, James Logan, Theodore Golfopoulos, Alexey Kaplan, Philip Michael, Richard Murray, Charlie Sanabria, Shane Schweiger, Thomas Toland, Rui Vieira \(1\)](#)
[Arrigoni, Luca Maria \(1\)](#)
[Jeon, Kibum \(1\)](#)
[Li, Kendrick Qijun \(1\)](#)
[Li, Xinyu \(1\)](#)

Publication Year: 2026 ✕

1 to 6 of 6 Results

Sort ▾



Machine Learning-Based Prediction of Bloodstream Infection in Rectal Carbapenem-**Resistant** Enterobacterales Carriers: A Multicenter Retrospective Cohort Study
Jan 7, 2026
Jeon, Kibum, 2026, "Machine Learning-Based Prediction of Bloodstream Infection in Rectal Carbapenem-Resistant Enterobacterales Carriers: A Multicenter Retrospective Cohort Study", <https://doi.org/10.7910/DVN/DQ2CBX>, Harvard Dataverse, V1
639 patients with 173 features



Replication Data for: Correcting Nonignorable Nonresponse Bias in Turnout Estimation Using Callback Data
Jan 14, 2026 - Political Analysis Dataverse
Li, Xinyu; Ying, Naiwen; Li, Kendrick Qijun; Shi, Xu; Miao, Wang, 2026, "Replication Data for: Correcting Nonignorable Nonresponse Bias in Turnout Estimation Using Callback Data", <https://doi.org/10.7910/DVN/TNABTF>, Harvard Dataverse, V1
... Based on the ANES Non-Response Follow-Up study concerning the 2020 U.S. presidential election, we investigate the role of callback data, i.e., records of contact attempts in the survey course, in adjusting for nonresponse bias in the estimation of turnout. We propose a stableness of **resistance** assumption to account for nonignorable missingness in the outcome, which states that the impact of the missing outcome on the response propensity is stable in the first two call attempts. Under this assumption and by integrating with covariates information from the census data, we establish identifiability and develop estimation methods for turnout. ...



Focus group data for: Claims to ignorance as a form of participation in transitional justice
Jan 10, 2026

[Harvard Dataverse](#)

MIMIC (Medical Information Mart for Intensive Care) Database

Medical Information Mart for Intensive Care III (MIMIC-III) is a large, freely-available database comprising de-identified health-related data associated with over 40,000 patients who stayed in critical care units of the Beth Israel Deaconess Medical Center between 2001 and 2012.

The database includes information such as demographics, vital sign measurements made at the bedside (~1 data point per hour), laboratory test results, procedures, medications, caregiver notes, imaging reports, and mortality (both in and out of hospital).

MIMIC-III supports a diverse range of analytic studies spanning epidemiology, clinical decision-rule improvement, and electronic tool development. It is notable for three factors:

- it is freely available to researchers worldwide
- it encompasses a diverse and very large population of ICU patients
- it contains high temporal resolution data including lab results, electronic documentation, and bedside monitor trends and waveforms

<https://browse.welch.jhmi.edu/datasets/ehr-databases>

Hopkins Initiatives

- COVID-19 Precision Medicine Analytics Platform Registry (JH-CROWN)

The main data source is Johns Hopkins' electronic medical record, Epic. The registry is refreshed weekly with new and updated data and is available for Johns Hopkins investigators to analyze subsets of the COVID-19 patient population for retrospective analyses. CHSOR members Dr. Jodi Segal and Dr. Caleb Alexander have been using these data.

- COVID-19-specific Common Data Model

PCORnet®, the National Patient-Centered Clinical Research Network, is creating a COVID-19-specific Common Data Model that will allow the use of information from patients across PCORnet's network. At Johns Hopkins, Dr. Harold Lehmann leads these activities.

- Johns Hopkins COVID-19 Collaboration Platform

Over 400 such trials have been registered on clinicaltrials.gov with dozens being added each day. Many of them are designed to answer similar questions and combining data or aggregating evidence could dramatically increase their efficiency and precision, getting answers to doctors faster and more reliably.
more...

- National COVID Cohort Collaborative

The National COVID Cohort Collaborative is the partnership among the NCATS-supported Clinical and Translational Science Awards (CTSA) Program hubs and the National Center for Data to Health (CD2H). At Johns Hopkins, Dr. Christopher Chute has taken the lead on this activity.

COVID-19 Datasets

- 4CE: Consortium for Clinical Characterization of COVID-19 by EHR
- Figshare: COVID-19 open data
- GitHub: COVID-19 Open Repo Data
- Harvard Dataverse: COVID-19 Data
- ICPSR: COVID-19 Data Repository
- ImmPort: COVID 19
- Mendeley Data: Elsevier COVID-19 Research Environment
- National COVID Cohort Collaborative (N3C)
OHDSI: Characterizing Health Associated Risks, and Your Baseline Disease In SARS-COV-2 (CHARYBDIS)
- Open-Access Data and Computational Resources to Address COVID-19.
- OpenSAFELY
- Tableau: COVID-19 Data Hub
- Vivli: Covid data
- Zenodo: Coronavirus Disease Research Community - COVID-19
- COVID-19 Research Database

Vivli Clinical Platform

[Home](#)[About](#)[Members](#)[News & Events](#)[Resources](#)[Portals](#)[LOG IN](#)

A global clinical research data sharing platform

The Vivli team is dedicated to helping researchers share and access data from clinical trials to advance science.

[SEARCH FOR STUDIES](#)[SUBMIT YOUR STUDY](#)

Vivli Clinical Platform



Sign up

Log In

KEYWORD SEARCH

BROWSE

What are you looking for today?



Antimicrobials

Select Multiple



Organisms

Select Multiple



Time of Sample

Select Multiple



Data Contributor

Select Multiple



Resistance Grouping

Select Multiple



Country

Select Multiple



Region

Select Multiple



Years Data Collected

From

To

Select One



Select One



Sources of Samples

Select Multiple



☐ Includes Genotype Information

☐ Includes Pediatrics Information

9

Datasets

Zenodo, Figshare Etc

Subjects

☒ COVID-19

☐ Coronavirus

☐ SARS-CoV-2

☐ coronavirus

☐ covid-19

☐ Covid-19

☐ covid

☐ Coronaviridae

☐ pathogens

☐ CETAF-taskforce

File type

☐ PDF

☐ BIN

☐ ZIP

Clear

March 16, 2020 (2020-05-01) Dataset Open

981

COVID-19 Open Research Dataset (CORD-19)

Sebastian Kohlmeier; Kyle Lo; Lucy Lu Wang

Important: This dataset is updated regularly and the latest version for download can be found here. In response to the COVID-19 pandemic, the Allen Institute for AI has partnered with leading research groups to prepare and distribute the COVID-19 Open Research Dataset (CORD-19), a free resource of scholarly articles, including full text conten...

Uploaded on May 7, 2020

7 more versions exist for this record

13243 7422

April 11, 2020 (v1) Preprint Open

141

Immunological mechanisms explaining the role of IgE, mast cells, histamine, elevating ferritin, IL-6, D-dimer, VEGF levels in COVID-19 and dengue, potential treatments such as mast cell stabilizers, antihistamines, Vitamin C, hydroxychloroquine, ivermectin and azithromycin

Arumugham, Vinu

A novel coronavirus, SARS-CoV-2 was identified in Wuhan, China. The disease caused by the virus can range in severity from asymptomatic to acute respiratory distress syndrome (ARDS) and death. Primary dengue infection results in IgE mediated sensitization against dengue virus (DENV) proteins. These IgE bind to receptors on mast cells. Upon sub...

Uploaded on April 11, 2020

13229 7445

November 13, 2020 (2020-10-20) Journal article Open

4,293

COVIPENDIUM: information available to support the development of medical countermeasures and interventions against COVID-19

Martine DENIS; Valerie VANDEWEERD; Rein VERBEEKE

<https://zenodo.org/search?q=coronavirus&f=subject%3ACOVID-19&l=list&p=1&s=10&sort=mostviewed>

Antibiotic Resistance Microbiology Dataset (ARMD): A resource for antimicrobial resistance from EHRs

[Nateghi Haredasht, Fateme](#) ¹  ; [Amrollahi, Fatemeh](#) ¹ ; [Maddali, Manoj](#) ¹ ; [Marshall, Nicholas](#) ¹ ; [Ma, Stephen](#) ¹ ; [Chang, Amy](#) ¹ ; [Banaei, Niaz](#) ¹ ; [Deresinski, Stanley](#) ¹ ; [Asch, Steven](#) ¹ ; [Goldstein, Mary](#) ¹ ; [Chen, Jonathan](#) ¹ 

[Author affiliations](#) ▼

Published Jan 23, 2025; Updated Oct 22, 2025 on Dryad. <https://doi.org/10.5061/dryad.jq2bvq8kp>

Data files

▼ Jan 23, 2025 version files	24.40 GB
▼ Mar 04, 2025 version files	24.48 GB
▼ Mar 04, 2025 version files	24.48 GB
▼ Apr 11, 2025 version files	23 GB
▼ Apr 11, 2025 version files	23 GB
▲ Oct 22, 2025 version files	23.77 GB 

Citation ▼



1923
views

1122
downloads

6
citations

Share:       

License: [Public domain](#)  

Subject keywords

[Antibiotic susceptibility testing](#),
[Antimicrobial resistance](#), [Infectious diseases](#), [Clinical medicine](#), [antibiotic stewardship](#), [EHR data](#), [Microbiological cultures](#)

Funding

[National Institute of Allergy and Infectious Diseases: GRANT13727773](#)

Related works

Primary article From: [Scientific Data](#)

 <https://doi.org/10.1038/s41597-025-0>



NICOLE WHEELER · UPDATED 6 YEARS AGO

76

<> Code

Download



Predicting antibiotic resistance in gonorrhoea

Detect antibiotic resistance using segments of the bacteria's DNA



Data Card

Code (14)

Discussion (1)

Suggestions (0)

About Dataset

Context

In this project, you will learn how to fit a model for predicting resistance in bacteria, and see how different forms of cross-validation impact the interpretation of your performance results.

We will be focussing on a species called *Neisseria gonorrhoeae*, bacteria which cause gonorrhoea. Gonorrhoea is the second most common sexually transmitted infection (STI) in Europe, after chlamydia. Rates of gonorrhoea infection are on the rise, with a 26% increase reported from 2017-2018 in the UK.

Many people who are infected (especially women) experience no symptoms, helping its spread. However if the infection is left untreated, it can lead to infertility in women, and can occasionally spread to other parts of the body such as your joints, heart valves, brain or spinal cord.

Usability

9.71

License

[CC0: Public Domain](#)

Expected update frequency

Never

Tags

Biology

Healthcare

Published May 30, 2018 | Version v1.0

Software

Open

SAFE-ICU/AMRsteward: stone

Tavpritesh; shubham14101; Tavpritesh@SAFE-ICU

Release version of AMR_Steward, the Bayesian Artificial Intelligence Dashboard for Stewarding Antimicrobial Stewardship at AIIMS, New Delhi, India.

Files

SAFE-ICU/AMRsteward-v1.0.zip

SAFE-ICU/AMRsteward-v1.0.zip

SAFE-ICU-AMRsteward-602c641

AMRsteward.Rproj	356 Bytes
DESCRIPTION	1.2 kB
LICENSE	35.1 kB

datasetsearch.research.google.com

datasetsearch.research.google.com/search?src=0&query=antibiotic%20resistance%20datasets&docid=L2cvMTF5YmZoYmduOQ%3D%3D

Google

antibiotic resistance datasets

▼ Last updated ▼ Download format Croissant ▼ Usage rights ▼ Topic ▼ Provider Free Saved datasets

100+ datasets found

kaggle Predicting antibiotic resistance in gonorrhoea
kaggle.com
zip
Updated Oct 5, 2019

D Antibiotic Resistance Microbiology Dataset (ARMD): A...
datadryad.org
zip
Updated Jan 23, 2025
+ more versions

E Antibiotic resistance and resistant genes in Escherichia coli from...
catalogue.ceh.ac.uk
hosted-metadata.bgs.ac.uk
+2more
zip

Antibiotic resistance and resistant genes in Escherichia coli from human, poultry and environmental samples in urban and rural Bangladesh (2017-18) [See More Versions](#)

Explore at: [Environmental Information D...](#) [hosted-metadata.bgs.ac.uk](#) [data-search.nerc.ac.uk](#) [data.europa.eu](#)

4 scholarly articles cite this dataset ([View in Google Scholar](#))

zip

Unique identifier
<https://doi.org/10.5285/0239cdaf-deab-4151-8f68-715063eaea45>

Dataset updated
Feb 8, 2019

Dataset provided by
NERC EDS Environmental Information Data Centre

Authors
E.K. Rousham; M.A. Islam; L. Unicomb; M. Asaduzzaman; R. Islam; B. Amin; T.I.M.A Mozmader; M.I. Hossain; M. Rahman; P. J. Wood

License
<https://eidc.ac.uk/licences/ogl/plain>



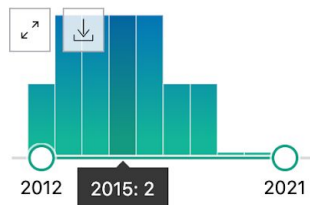

[Search](#)
[Advanced](#) [Create alert](#) [Create RSS](#)
[User Guide](#)
[Save](#)
[Email](#)
[Send to](#)

Sorted by: Best match

[Display options](#)

MY NCBI FILTERS 

RESULTS BY YEAR



TEXT AVAILABILITY

☐ Abstract

☐ Full Text

6 results


[Database resources of the National Center for Biotechnology Information.](#)

1

NCBI Resource Coordinators.

Cite

Nucleic Acids Res. 2018 Jan 4;46(D1):D8-D13. doi: 10.1093/nar/gkx1095.

PMID: 29140470 [Free PMC article.](#)

Share


[Database resources of the National Center for Biotechnology Information.](#)

2

NCBI Resource Coordinators.

Cite

Nucleic Acids Res. 2016 Jan 4;44(D1):D7-19. doi: 10.1093/nar/gkv1290. Epub 2015 Nov 28.

PMID: 26615191 [Free PMC article.](#)

Share

Database	Description	URL
NCBI GenBank	Genetic sequence database	www.ncbi.nlm.nih.gov/genbank
Ensembl	A genome browser of vertebrates	www.ensembl.org
NCBI Refseq	A collection of non-redundant and well-annotated genomic sequences, transcripts and proteins	www.ncbi.nlm.nih.gov/refseq
UCSC genome browser	Interactive genome visualization browser	https://genome.ucsc.edu/
1000 genomes	A catalogue of human genomic variation	www.internationalgenome.org
GeneCards	An integrative database of predicted and annotated human genes	www.genecards.org
lncRNAdb	Database containing annotated long non-coding RNAs in eukaryotes	rnacentral.org/expert-database/lncrnadb
miRBase	Database of published miRNA sequences along with annotation	www.mirbase.org
DIANA-TarBase	A database of experimentally supported miRNA targets.	www.microrna.gr/tarbase